

UNIT V

LASERS

INTRODUCTION:

LASER stands for Light Amplification by Stimulated Emission of Radiation. Laser technology started with Albert Einstein in 1917, he has given theoretical basis for the development of Laser. The technology further evolved in 1960 when the very first laser called Ruby Laser was built at Hughes Research Laboratories by T.H. Mainmann.

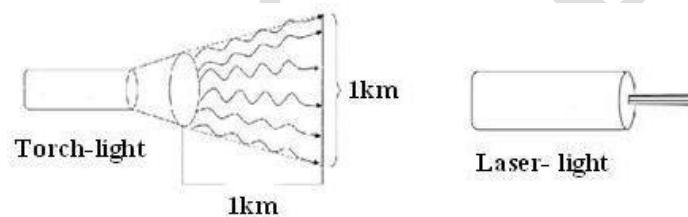
CHARACTERISTIC OF LASER RADIATION:

The laser beam has the properties given below which distinguish it from an ordinary beam of light. Those are

1. Highly directional
2. Highly monochromatic
3. Highly intense
4. Highly coherence

1. Highly directional:

A conventional light source emits light in all directions. On the other hand, Laser emits light only in one direction. The width of Laser beam is extremely narrow and hence a laser beam can travel to long distances without spreading.



The directionality of laser beam is expressed in terms of divergence

$$\Delta\theta = \frac{r_2 - r_1}{d_2 - d_1}$$

Where r_1 and r_2 are the radii of laser beam spots at distances of d_1 and d_2 respectively from laser source.

2. Highly monochromatic:

A monochromatic source is a single frequency or single wavelength source of light. The laser light is more monochromatic than that of a convectional light source. This may be due to the stimulated characteristic of laser light. The band width of convectional monochromatic light source is $1000\text{\AA}$. But the band width of ordinary light source is $10\text{\AA}$. For high sensitive laser source is $10^{-8}\text{\AA}$.

3. Highly intense:

Laser light is highly intense than the conventional light. A one milli-Watt He-Ne laser is highly intense than the sun intensity. This is because of coherence and directionality of laser. Suppose when two photons each of amplitude 'A' are in phase with other, then young's principle of superposition, the resultant amplitude of two photons is $2A$ and the intensity is $4a^2$. Since in laser many numbers of photons are in phase with each other, the amplitude of the resulting wave becomes 'nA' and hence the intensity of laser is proportional to n^2A^2 . So 1mw He-Ne laser is highly intense than the sun.

4. Highly coherence

A predictable correlation of the amplitude and phase at any one point with other point is called coherence. In case of conventional light, the property of coherence exhibits between a source and its virtual source whereas in case of laser the property coherence exists between any two sources of same phase.

There are two types of coherence

- i) Temporal coherence
- ii) Spatial coherence.

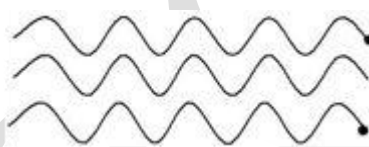
Temporal coherence (or longitudinal coherence):

The predictable correlation of amplitude and phase at one point on the wave train w.r.t. another point on the same wave train, then the wave is said to be temporal coherence.



Spatial coherence (or transverse coherence):

The predictable correlation of amplitude and phase at one point on the wave train w.r.t. another point on a second wave, then the waves are said to be spatial coherence (or transverse coherence). Two waves are said to be coherent when the waves must have same phase & amplitude.



INTERACTION OF LIGHT WITH MATTER AND THE THREE QUANTUM PROCESSES:

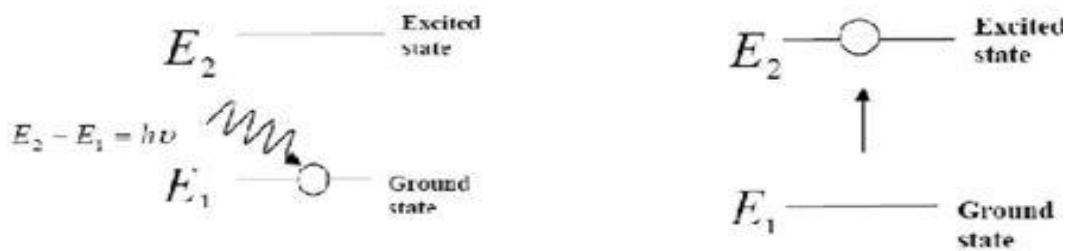
When the radiation interacts with matter, results in the following three important phenomena. They are

- (i) Induced or Stimulated Absorption
- (ii) Spontaneous Emission
- (iii) Stimulated Emission

STIMULATED ABSORPTION (OR) INDUCED ABSORPTION (OR) ABSORPTION:

An atom in the lower energy level or ground state energy level (E_1) absorbs the incident photon and goes to excited state (E_2) as shown in figure below. This process is called induced or stimulated absorption.

Let E_1 and E_2 be the energies of ground and excited states of an atom. Suppose, if a photon of energy $E_2 - E_1 = h\nu$ interacts with an atom present in the ground state, the atom gets excitation from ground state E_1 to excited state E_2 . This process is called stimulated absorption.



Stimulated absorption rate depends upon the number of atoms available in the lowest energy state as well as the energy density of photons.

SPONTANEOUS EMISSION:

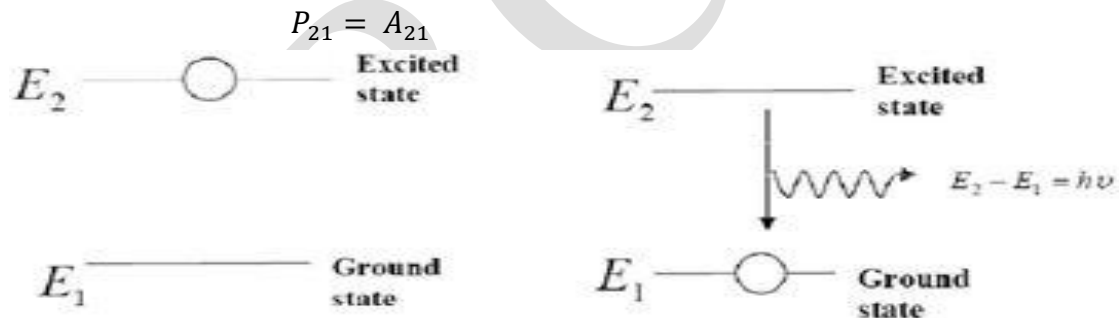
The atom in the excited state returns to ground state emitting a photon of energy $(E) = E_2 - E_1 = h\nu$, without applying an external energy spontaneously is known as spontaneous emission.

Let E_1 and E_2 be the energies of ground and excited states of an atom. Suppose, if photon of energy $E_2 - E_1 = h\nu$ interacts with an atom present in the ground state, the atom gets excitation from ground state E_1 to excited state E_2 .

The excited atom does not stay for a long time in the excited state. The excited atom gets de-excitation after its life time by emitting a photon of energy $E_2 - E_1 = h\nu$. This process is called spontaneous emission.

The spontaneous emission rate depends up on the number of atoms present in the excited state.

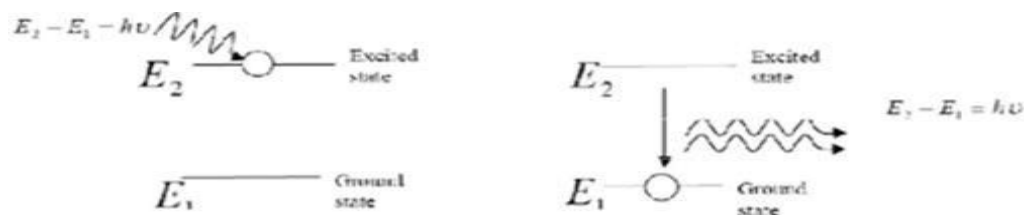
The probability of spontaneous emission (P_{21}) is independent of $u(\nu)$.



STIMULATED-EMISSION:

The atom in the excited state can also returns to the ground state by applying external energy or inducement of photon thereby emitting two photons which are having same energy as that of incident photon. This process is called as stimulated emission.

Stimulated emission was postulated by Einstein. Let E_1 and E_2 be the energies of ground and excited states of an atom. Let a Photon of energy $E_2 - E_1 = h\nu$ interacts with the excited atom with in their life time The atom gets de-excitation to ground state by emitting of another photon. These photons have same phase and it follows coherence. This phenomenon is called stimulated emission



Stimulated emission rate depends upon the number of atoms available in the excited state as well as the energy density of photons.

Comparison between Spontaneous and Stimulated emission:

Spontaneous emission	Stimulated emission
1. The spontaneous emission was Postulated by Bohr.	1. The stimulated emission was Postulated by Einstein.
2. Additional photons are not required in spontaneous emission.	2. Additional photons are required in Stimulated emission.
3. One Photon is emitted in spontaneous emission.	3. Two photons are emitted in stimulated emission.
4. The emitted radiation is incoherent.	5. The emitted radiation is coherent.
5. The emitted radiation is less intense.	6. The emitted radiation is high intense.

Light Amplification:

Light amplification requires stimulated emission exclusively. In practice, absorption and spontaneous emission always occur together with stimulated emission. The laser operation is achieved when stimulated emission exceeds the other two processes due to its higher transitions rates of atomic energy levels.

EINSTEIN COEFFICIENTS & THEIR RELATIONS

Einstein coefficients:

Einstein mathematically expressed the statistical nature of the three possible radiative transition routes (spontaneous emission, stimulated emission, and absorption) with the so-called Einstein coefficients and quantified the relations between the three processes.

Let N_1 be the number of atoms per unit volume with energy E_1 and N_2 be the number of atoms per unit volume with energy E_2 . Let ' n ' be the number of photons per unit volume at frequency ' ν ' such that $E_2 - E_1 = h\nu$. Then, the energy density of photons $E = h\nu$.

- **Stimulated absorption:** When the photons interact with the atoms it leads to absorption transition which is called as stimulated absorption. Stimulated absorption rate depends upon the number of atoms available in the lowest energy state as well as the energy density photons.

$$\begin{aligned}\text{Stimulated absorption rate} &\propto N_1 \\ &\propto u(\nu) \\ &= B_{12} N_1 u(\nu)\end{aligned}$$

Where B_{12} is the Einstein coefficient of stimulated absorption.

- **Spontaneous emission:** The atom in the excited state returns to ground state emitting a photon of energy $(E) = E_2 - E_1 = h\nu$, without applying an external energy spontaneously is known as spontaneous emission. The spontaneous emission rate depends up on the number of atoms present in the excited state.

$$\begin{aligned}\text{Spontaneous emission rate} &\propto N_2 \\ &= A_{21} N_2\end{aligned}$$

Where A_{21} is the Einstein coefficient of spontaneous emission.

- **Stimulated emission:** The atom in the excited state can also returns to the ground state by applying external energy or inducement of photon thereby emitting two photons which are having same energy as that of incident photon. This process is called as stimulated emission. Stimulated emission rate depends upon the number of atoms available in the excited state as well as the energy density of incident photons.

$$\begin{aligned}\text{Stimulated emission rate} &\propto N_2 \\ &\propto u(\nu) \\ &= B_{21} N_2 u(\nu)\end{aligned}$$

Where B_{21} is the Einstein coefficient of stimulated emission.

If the system is in equilibrium, the rates of absorption transitions (Stimulated absorption) are equal to emission transitions (The rate of Spontaneous + Stimulated emission).

The rate of Absorption = The rate of Spontaneous + Stimulated emission

$$\begin{aligned}B_{12}N_1u(\nu) &= A_{21}N_2 + B_{21}N_2u(\nu) \\ B_{12}N_1u(\nu) - B_{21}N_2u(\nu) &= A_{21}N_2 \\ u(\nu)(B_{12}N_1 - B_{21}N_2) &= A_{21}N_2 \\ u(\nu) &= \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2} \\ u(\nu) &= \left(\frac{A_{21}}{B_{21}} \times \frac{1}{\left[\frac{N_1}{N_2} \times \frac{B_{12}}{B_{21}} - 1 \right]} \right)\end{aligned}$$

According to Boltzmann distribution law

$$\begin{aligned}N &= N_0 e^{-\frac{E}{k_B T}} \\ N_1 &= N_0 e^{-\frac{E_1}{k_B T}} \\ N_2 &= N_0 e^{-\frac{E_2}{k_B T}}\end{aligned}$$

$$\frac{N_1}{N_2} = \frac{N_0 e^{-\frac{E_1}{k_B T}}}{N_0 e^{-\frac{E_2}{k_B T}}} = e^{\frac{E_2 - E_1}{k_B T}} = e^{\frac{h\nu}{k_B T}}$$

$$u(\nu) = \frac{A_{21}}{B_{21}} \times \frac{1}{\left[e^{\frac{h\nu}{k_B T}} \left(\frac{B_{12}}{B_{21}} \right) - 1 \right]} \quad \text{---(1)}$$

According to Planck's radiation formula the energy density of photon is given by,

$$u(\nu) = \frac{8\pi h\nu^3}{c^3} \times \left(\frac{1}{\left[e^{\frac{h\nu}{k_B T}} - 1 \right]} \right) \quad \text{---(2)}$$

Comparing eq.(1) & eq.(2), we get

$$(i) \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad \text{or} \quad \frac{A_{21}}{B_{21}} \propto \nu^3$$

$$(ii) \quad \frac{B_{12}}{B_{21}} = 1 \quad \text{or,} \quad B_{12} = B_{21}$$

POPULATION INVERSION:

The number of atoms present in the excited (or higher) state is greater than the number of atoms present in the ground energy state (or lower state) is called population inversion.

Let us consider two level energy systems of energies E_1 and E_2 as shown in figure. Let N_1 and N_2 be the population (means number of atoms per unit volume) of E_1 and E_2 respectively.

According to Boltzmann's distribution the population of an energy level E , at temperature T is given by

$$N_i = N_0 e^{\left(-\frac{E_i}{k_B T}\right)} \quad \text{where } i = 1, 2, 3, \dots, N_i$$

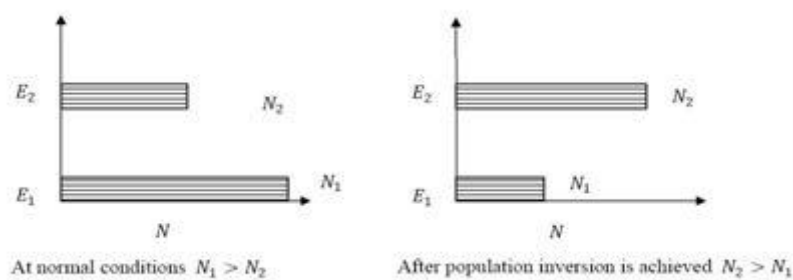
where ' N_0 ' is the number of atoms in ground or lower energy states & k is the Boltzmann constant.

From the above equation the population of energy levels E_1 & E_2 are given by

$$N_1 = N_0 e^{\left(-\frac{E_1}{k_B T}\right)}$$

$$N_2 = N_0 e^{\left(-\frac{E_2}{k_B T}\right)}$$

At ordinary conditions $N_1 > N_2$ i.e., the population in the ground or lower state is always greater than the population in the excited or higher states. The stage of making, population of higher energy level greater than the population of lower energy level is called population inversion i.e., $N_2 > N_1$.



META STABLE STATE:

In general the number of excited particles in a system is smaller than the non excited particles. The time during which a particle can exist in the ground state is unlimited. On the other hand, the particle can remain in the excited state for a limited time known as life time.

The life time of the excited hydrogen atom is of the order of 10^{-8} sec. However there exist such excited states in which the life time is greater than 10^{-8} sec. These states are called as Meta stable states.

PUMPING MECHANISMS (OR TECHNIQUES):

Pumping:

The process of rising more no of atoms to the excited state by artificial means is called pumping.

A system in which population inversion is achieved is called as an active system. The method of raising the particles from lower energy state to higher energy state is called pumping. (or the process of achieving of population inversion is called pumping). This can be done by number of ways. The most commonly used pumping methods are

- Optical pumping
- Electrical discharge pumping
- Chemical pumping
- Injection current pumping

Optical pumping:

Optical pumping is used in solid laser. Xenon flash tubes are used for optical pumping. Since these materials have very broad band absorption, sufficient amount of energy is absorbed from the emission band of flash lamp and population inversion is created. Examples of optically pumped lasers are ruby, Nd: YAG Laser ($Y_3 Al_5 G_{12}$) (Neodymium: Yttrium Aluminum Garnet), Nd: Glass Laser

Electrical discharge pumping:

Electrical discharge pumping is used in gas lasers. Since gas lasers have very narrow absorption band pumping them any flash lamp is not possible. Examples of Electrical discharge pumped lasers are He-Ne laser, CO_2 laser, argon-ion laser, etc

Chemical pumping:

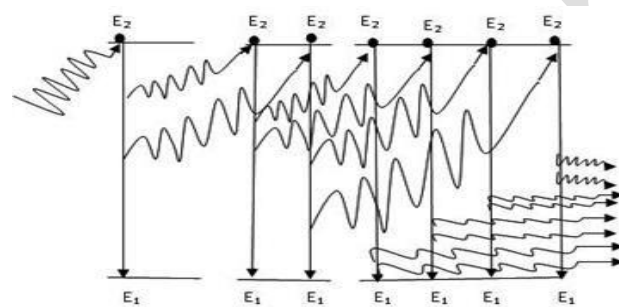
Chemical reaction may also result in excitation and hence creation of population inversion in few systems. Examples of such systems are HF and DF lasers.

Injection current pumping:

In semiconductors, injection of current through the junction results in creates of population inversion among the minority charge carriers. Examples of such systems are InP and GaAs.

PRINCIPLE OF LASER/LASING ACTION:

Let us consider many no of atoms in the excited state. Now the stimulating photon interacts with any one of the atoms in the excited state, the stimulated emission will occur. It emits two photons, having same energy & same frequency move in the same direction. These two photons will interact with another two atoms in excited state & emit 8-photons. In a similar way chain reaction is produced this phenomenon is called “Principle of lasing –action”. We get a monochromatic, coherent, directional & intense beam is obtained. This is called laser beam. This is the principle of working of a laser.

**Components of a LASER:**

Any laser system consists of 3-important components. They are

- (i) Source of energy or pumping source
- (ii) Active-medium (Laser Material)
- (iii) Optical cavity or resonator

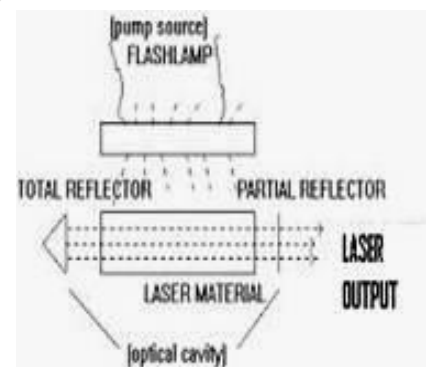
(i) Energy Source : It supply energies & pumps the atoms or molecules in the active medium to excited states. As a result we get population inversion in the active medium which emits laser.

Ex: Xenon flash lamp, electric field.

(ii) Active medium: The medium in which the population inversion takes place is called as active medium.

Active-centre: The material in which the atoms are raised to excited state to achieve population inversion is called as active center.

(iii) Optical-cavity or resonator: The active medium is enclosed between a fully reflected mirror & a partially reflective mirror. This arrangement is called as cavity or resonator. As a result, we get highly intense monochromatic, coherence laser light through the non-reflecting portion of the mirror.



DIFFERENT TYPES OF LASERS

On the basis of active medium used in the laser systems, lasers are classified into several types

- | | |
|-------------------------|--|
| I. Solid lasers | : Ruby laser, Nd:YAG laser, Nd:Glass |
| II. Liquid lasers | : Europium Chelate laser, SeOCl_2 |
| III. Gas lasers | : CO_2 , He-Ne, Argon-Ion Laser |
| IV. Dye lasers | : Rhodamine 6G |
| V. Semiconductor lasers | : InP, GaAs. |

RUBY LASER

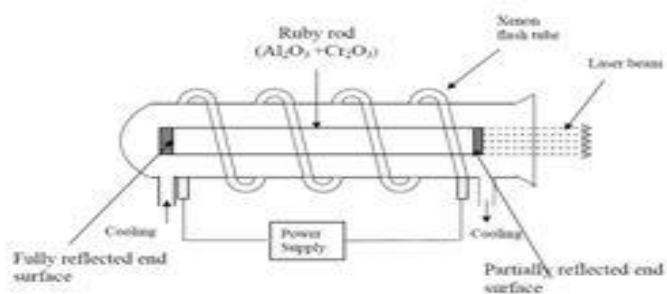
Ruby laser is a three level solid state laser and was developed by Mainmann in 1960. Ruby ($\text{Al}_2\text{O}_3 + \text{Cr}_2\text{O}_3$) is a crystal of Aluminium oxide, in which 0.05% of Al^{+3} ions are replaced by the Cr^{+3} ions. The colour of the ruby rod is pink. The active medium in the ruby rod is Cr^{+3} ions.

Principle or Characteristics of a ruby laser:

Due to optical pumping, the chromium atoms are raised to excited states then the atoms come to metastable state by non-radiative transition. Due to stimulated emission the transition of atoms takes place from metastable state to ground state and gives a laser beam.

Construction:

- In ruby laser 4cm length and 5mm diameter rod is generally used.
- Both the ends of the rods are highly polished and made strictly parallel.
- The ends are silvered in such a way, one becomes partially reflected and the other end fully reflected.
- The ruby rod is surrounded by xenon flash tube, which provides the pumping light to excite the chromium ions in to upper energy levels.

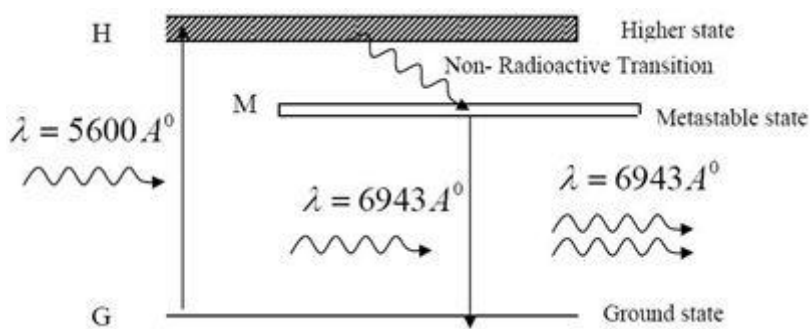


- Xenon flash tube emits thousands joules of energy in few milli seconds, but only a part of that energy is utilized by the chromium ions while the rest energy heats up the apparatus.
- A cooling arrangement is provided to keep the experimental set up at normal temperatures.

Working:

- The energy level diagram of chromium ions is shown in figure.
- The chromium ions get excitation into higher energy levels by absorbing of $5600\text{\AA}$ of wave length radiation.
- The excited chromium ions stay in the level H for short interval of time (10^{-8} Sec).
- After their life time most of the chromium ions are de-excited from H to G and a few

chromium ions are de-excited from H to M.



- The transition between H and M is non-radioactive transition i.e. the chromium ions give their energy to the lattice in the form of heat.
- In the Meta stable state the life time of chromium ions is 10^{-3} sec.
- Due to the continuous working of flash lamp, the chromium ions are excited to higher state H and returned to M level.
- After few milli seconds the level M is more populated than the level G and hence the desired population inversion is achieved.
- The state of population inversion is not a stable one.
- The process of spontaneous transition is very high.
- When the excited chromium ion passes spontaneously from H to M it emits one photon of wave length $6943\text{\AA}^0$.
- The photon reflects back and forth by the silver ends and until it stimulates an excited chromium ion in M state and it to emit fresh photon in phase with the earlier photon.
- The process is repeated again and again until the laser beam intensity is reached to a sufficient value.
- When the photon beam becomes sufficient intense, it emerges through the partially silvered end of the rod.
- The wave length $6943\text{\AA}^0$ is in the red region of the visible spectrum on returning to ground state (G).

Uses of Ruby laser:

- Used in distance measurement using 'pulse echo technique'
- Used for measurement of plasma properties such as electron density and temperature.
- Used to remove the melanin of the skin.
- Used for recording pulsed holograms.
- Used as target designators and range finders in military.

Draw backs of Ruby laser:

- It requires high pumping power.
- The efficiency of ruby laser is very small. It is a pulse laser.

He-Ne LASER

It was discovered by A. Javan & his co-workers in 1960. It is a continuous wave gas laser. It consists of mixture of He & Ne in 10:1 ratio as a active medium.

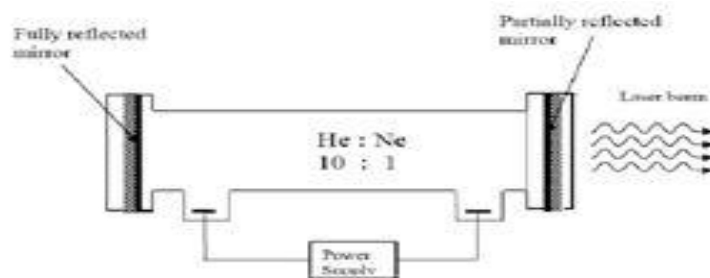
Principle/Characteristics of He-Ne laser:

This laser is based on the principle of stimulated emission, produced in the He & Ne. The

population inversion is achieved due to the interaction between He & Ne gases. Using gas lasers, we can achieve highly coherent, directional and high monochromatic beam.

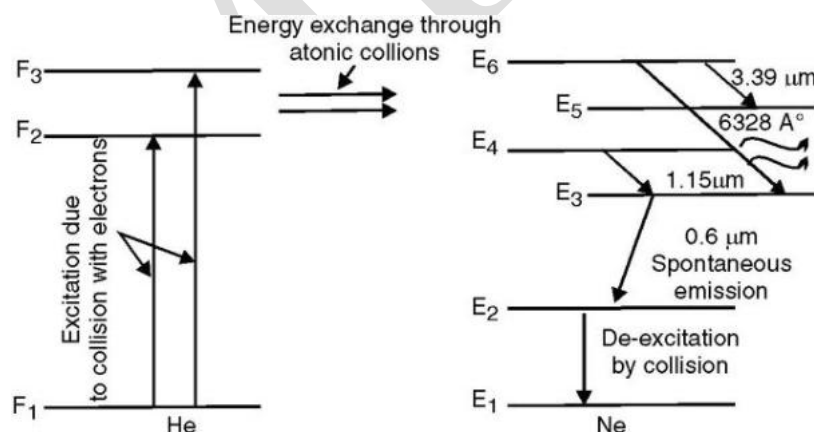
Construction:

- In He-Ne gas laser, the He and Ne gases are taken in the ratio 10:1 in the discharge tube.
- Two reflecting mirrors are fixed on either ends of the discharge tube, in that, one is partially reflecting and the other is fully reflecting.
- In He-Ne laser 80cm length and 1cm diameter discharge tube is generally used.
- The output power of these lasers depends on the length of the discharge tube and pressure of the gas mixture.
- Energy source of laser is provided by an electrical discharge of around 1000V through an anode and cathode at each end of the glass tube.



Working:

- When the electric discharge is passing through the gas mixture, the electrons accelerated towards the positive electrode.
- During their passage, they collide with He atoms and excite them into higher levels.
- F_2 and F_3 form F_1 . In higher levels F_2 and F_3 , the life time of He atoms is more.
- So there is a maximum possibility of energy transfer between He and Ne atoms through atomic collisions.
- When He atoms present in the levels F_2 and F_3 collide with Ne atoms present ground state E_1 , the Ne atoms gets excitation into higher levels E_4 and E_6 .



- Due to the continuous excitation of Ne atoms, we can achieve the population inversion between the higher levels E_4 (E_6) and lower levels E_3 (E_5).
- The various transitions $E_6 \rightarrow E_5$, $E_4 \rightarrow E_3$ and $E_6 \rightarrow E_3$ leads to the emission of wavelengths 3.39A° , 1.15A° and 6328A° .
- The first two corresponding to the infrared region while the last wavelength is

corresponding to the visible region.

- The Ne atoms present in the E_3 level are de-excited into E_2 level, by spontaneously emission of photon.
- When a narrow discharge tube is used, the Ne atoms present in the level E_2 collide with the walls of the tube and get de-excited to ground level E_1 .

Uses of He-Ne laser:

- Used in laboratories for all interferometric experiments.
- Used widely in metrology in surveying, alignment etc.
- Used to read barcodes and He-Ne laser scanners also used for optical character recognition.
- Used in holography.

APPLICATIONS OF LASERS

Due to high intensity, high mono-chromaticity and high directionality of lasers, they are widely used in various fields like

1. communication
2. computers
3. chemistry
4. photography
5. industry
6. medicine
7. military
8. scientific research

1. Communication:

- In case of optical communication, semiconductor laser diodes are used as optical sources.
- More channels can be sent simultaneously. Signal cannot be tapped as the bandwidth is large, more data can be sent.
- A laser is highly directional and has less divergence, hence it has greater potential use in space crafts and submarines.

2. Computers :

- In LAN (local area network), data can be transferred from memory storage of one computer to other computer using laser for short time.
- Lasers are used in CD-ROMS during recording and reading the data.

3. Chemistry :

- Lasers are used in molecular structure identification.
- Lasers are also used to accelerate some chemical reactions.
- Using lasers, new chemical compounds can be created by breaking bonds between atoms or molecules.

4. Photography :

- Lasers can be used to get 3-D lensless photography.
- Lasers are also used in the construction of holograms.

5. Industry :

- Lasers can be used to blast holes in diamonds and hard steel.
- Lasers are also used as a source of intense heat.
- Carbon dioxide laser is used for cutting drilling of metals and nonmetals, such as ceramics plastics, glass etc.
- High power lasers are used to weld or melt any material.
- Lasers are also used to cut teeth in saws and test the quality of fabric.

6. Medicine :

- Pulsed neodymium laser is employed in the treatment of liver cancer.
- Argon and carbon dioxide lasers are used in the treatment of liver and lungs.
- Lasers used in the treatment of Glaucoma.

7. Military :

- Lasers can be used as a war weapon.
- High energy lasers are used to destroy the enemy air-crafts and missiles.
- Lasers can be used in the detection and ranging like RADAR.

8. Scientific research:

- Lasers are used in the field of 3D-photography.
- Lasers used in Recording and reconstruction of hologram.
- Lasers are employed to create plasma.
- Lasers used to produce certain chemical reactions.
- Lasers are used in Raman spectroscopy to identify the structure of the molecule.
- Lasers are used in the Michelson- Morley experiment.
- A laser beam is used to confirm Doppler shifts in frequency for moving objects.